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Course code: CSA0302

Course Name:Data Structure

Q1:Reversed the number

Write a C program that takes an array as input and returns it reversed.

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,temp;
5
6     scanf("%d",&n);
7
8     int arr[n];
9     for(i=0;i<n;i++){
10         scanf("%d",&arr[i]);
11     }
12     for(i=0;i<n/2;i++){
13         temp=arr[i];
14         arr[i]=arr[n-1-i];
15         arr[n-1-i]=temp;
16     }
17     for(i=0;i<n;i++){
18         printf("%d",arr[i]);
19     }
20     return 0;
21
22 }
```

Input

5
1 2 3 4 5

Output

54321

Q2:Program for palindrom

Write a C program for palindrom

Your Code

```
1 #include<stdio.h>
2 #include<string.h>
3 int main() {
4     char str[100];
5     int i,len, flag=1;
6
7     scanf("%d",str);
8
9     len=strlen(str);
10
11     for(i=0;i<len/2;i++){
12         if(str[i]!=str[len-i-1]){
13             flag =0;
14             break;
15         }
16     }
17     if(flag)
18         printf("palindrome");
19     else
20         printf("not palindrome");
21     return 0;
22 }
```

Input

hello

Output

not palindrome

Q3:Searching an element in Array

```

1 #include<stdio.h>
2 int main(){
3     int n,arr[100],key;
4     int low=0,high,mid;
5
6     scanf("%d",&n);
7
8     for(int i=0;i<n;i++){
9         scanf("%d",&arr[i]);
10    }
11
12    scanf("%d",&key);
13
14    high = n - 1;
15    while(low<=high){
16        mid=(low+high)/2;
17
18        if(arr[mid]==key){
19            printf("Element found at index %d",mid);
20            return 0;
21        }
22        else if(arr[mid]<key)
23            low=mid+1;
24        else
25            high=mid-1;
26    }
27    printf("element not found");
28    return 0;
29 }

```

```

5
10 20 30 40 50
30

```

Output



Q4:Occurrences of an Element

Write a C program for Count the Occurrences of an Element

Your Code

```

1 #include<stdio.h>
2 int main(){
3     int n,arr[100],key,count=0;
4
5     scanf("%d",&n);
6
7     for(int i=0;i<n;i++){
8         scanf("%d",&arr[i]);
9     }
10    scanf("%d",&key);
11
12    for(int i=0;i<n;i++){
13        if(arr[i]==key)
14            count++;
15    }
16    printf("%d",count);
17    return 0;
18 }

```

Input

```

5
1 2 3 2 2
2

```

Output

```

3

```

Q5: Binary Search

```
1 #include<stdio.h>
2 int main() {
3     int n,arr[100],key;
4     int low=0,high,mid;
5
6     scanf("%d",&n);
7
8     for(int i=0;i<n;i++)
9         scanf("%d",&arr[i]);
10
11     scanf("%d",&key);
12
13     high =n-1;
14     while(low<=high){
15         mid=(low+high)/2;
16
17         if(arr[mid]== key){
18             printf("Element found at index%d",mid);
19             return 0;
20         }
21         else if (arr[mid]<key)
22             low =mid+1;
23         else
24             high = mid-1;
25     }
26     printf("element not found");
27     return 0;
28 }
```

```
5
10 20 30 40 50
40
```

Output

```
Element found at index3
```

Q6: Merge Two Arrays

```
1 #include<stdio.h>
2 int main() {
3     int n1,n2,arr1[100],arr2[100],arr3[200];
4     int i,j;
5
6     scanf("%d",&n1);
7
8     for(int i=0;i<n1;i++)
9         scanf("%d",&arr1[i]);
10
11     scanf("%d",&n2);
12
13     for(int i=0;i<n2;i++)
14         scanf("%d",&arr2[i]);
15
16     for(int i=0;i<n1;i++)
17         arr3[i]=arr1[i];
18
19     for(int j=0;j<n2;j++)
20         arr3[i++]=arr2[j];
21
22     for(int i=0;i<n1+n2;i++)
23         printf("%d",arr3[i]);
24
25     return 0;
26 }
```

```
3
1 2 3
3
4 5 6
```

Output

```
456000
```

Q7:Largest and Smallest Element

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int n,a[100],i,max,min;
4
5     scanf("%d",&n);
6
7     for(i=0;i<n;i++){
8         scanf("%d",&a[i]);
9     }
10    max=min=a[0];
11
12    for(i=1;i<n;i++){
13        if(a[i]>max)max=a[i];
14        if(a[i]<min)min=a[i];
15    }
16    printf("%d\n%d",max,min);
17    return 0;
18 }
```

Input

```
5
10 25 8 40 15
```

Output

```
40
8
```

Q8:Second Largest Element

Write a C program for Find the Second Largest Element

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int n,a[100],i,max,second;
4
5     scanf("%d",&a[i]);
6     for(i=0;i<n;i++){
7         scanf("%d",&a[i]);
8     }
9     max = second = -9999;
10
11    for(i=0;i<n;i++){
12        if(a[i]>max){
13            second =max;
14            max=a[i];
15        }else if(a[i] >second && a[i] !=max){
16            second =a[i];
17        }
18    }
19    printf("%d",second);
20    return 0;
21 }
```

Input

```
5
10 20 40 30 15
```

Output

Q9:First and Last Occurrence of an Element

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int n,a[100],x,i,first=-1,last=-1;
4
5     scanf("%d",&n);
6
7     for(i=0;i<n;i++){
8         scanf("%d",&a[i]);
9     }
10    scanf("%d",&x);
11
12    for(i=0;i<n;i++){
13        if(a[i]==x){
14            if(first == -1)
15                first=i;
16            last = i;
17        }
18    }
19
20    printf("%d%d",first,last);
21    return 0;
22 }
23 }
```

Input

```
6
2 4 5 4 7 4
4
```

Output

```
15
```

Q10:Missing Element in Array

Write a C program for Find Missing Element in an Array

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int n,arr[100];
4     int i,sum=0,total;
5
6     scanf("%d",&n);
7
8     for(i=0;i<n-1;i++){
9         scanf("%d",&arr[i]);
10        sum+=arr[i];
11    }
12
13    total =(n*(n+1))/2;
14    printf("%d",total -sum);
15    return 0;
16 }
```

Input

```
5
1 2 3 5
```

Output

```
4
```